Last time we saw that the wave speed for transverse waves in a taut string is given by

$$v = \sqrt{\frac{T}{\mu}}$$

T = string tension

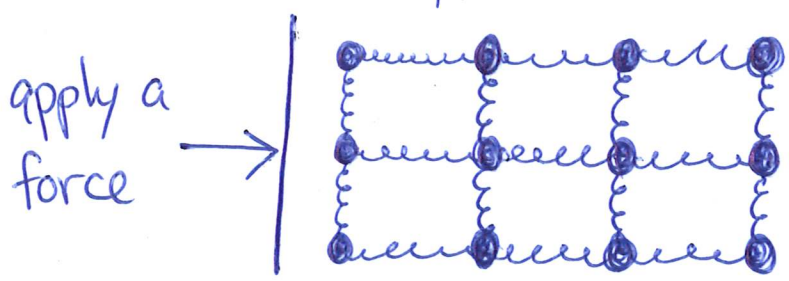
μ = linear mass density

Today we will be looking at longitudinal waves in solids and fluid media.

⇒ Slides showing transverse and longitudinal waves.

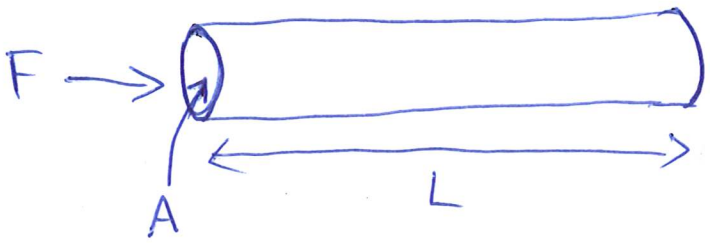
Let's start by considering a metal rod and zooming in to see the atoms that make it up. The atoms are arranged in a crystal lattice with nearest neighbours held in place by chemical bonds that act like springs

2D picture:



applying a force causes the spring bonds to compress and sends a compression wave through the lattice

Zoomed out 3D picture:



Apply a force F over an area A and it will cause the length L to change

Linear regime:

$F = -k\,\Delta L$ (Hooke's Law)

We define Young's modulus $Y \equiv \left| \frac{F/A}{\Delta L/L} \right|$ so that the effective spring constant is $k = Y\frac{A}{L}$. Young's modulus is a property of the material itself (related to the molecular bond strength), so it is much more useful than the "spring constant" of a particular rod, which is a geometric quantity.

The units of Y are: $\frac{N/m^2}{m/m} = \boxed{N/m^2 = Pa}$

The wave speed for longitudinal waves in a solid of density ρ and Young's modulus Y is

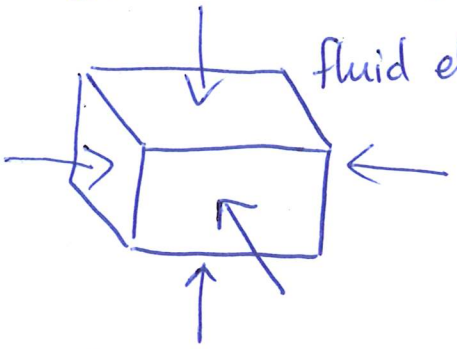
$$\boxed{v = \sqrt{\frac{Y}{\rho}}}$$ (compare with $v = \sqrt{\frac{T}{\mu}}$)

⇒ Top Hat Question

In a fluid, the atoms are not chemically bonded together but they are constantly in motion such that on average the pressure is equalized when the fluid is in equilibrium. The atoms don't know what their neighbours are doing, but collectively they prefer to be evenly distributed.

If the pressure increases, the volume decreases, and vice versa. The quantity that describes this elastic response of a fluid is called the

Bulk Modulus



bulk modulus describes how "squishy" the fluid is.

$$B \equiv \frac{-\Delta P}{\Delta V / V}$$

i.e. what is the response of the fluid to a change in pressure?

Units of Bulk modulus: $\frac{N/m^2}{m^3/m^3} = N/m^2 = Pa$

Same units as Young's modulus but different origin.

The wave speed in a fluid with bulk modulus B is

$$\boxed{v = \sqrt{\frac{B}{\rho}}}$$ (compare to $v = \sqrt{\frac{Y}{\rho}}$ and $v = \sqrt{\frac{T}{\mu}}$)

The bulk moduli and densities of various liquids can be looked up in tables. These are temperature-dependent quantities, so care must be taken when high accuracy is needed.

For gases we can analyze this further because we can make use of what we know about ideal gases.

For an adiabatic wave (i.e. reversible thermodynamic process, which will be true of any waves we consider)

$$PV^\gamma = \text{constant} \qquad \text{where} \quad \gamma = \frac{C_P}{C_V} = \begin{cases} 5/3 & \text{monatomic gases (He, Ar)} \\ 7/5 & \text{diatomic gases } (H_2, O_2, N_2, \text{etc}) \end{cases}$$

Consider taking Δ of both sides:

$$\Delta(PV^\gamma) = \Delta(\text{constant})$$

$$(\Delta P)V^\gamma + P(\gamma V^{\gamma-1}\Delta V) = 0$$

$$\left(\Delta P + P\gamma\frac{\Delta V}{V}\right)V^\gamma = 0$$

$$\Rightarrow \gamma P = -\frac{\Delta P}{\Delta V/V} \equiv B \quad \text{so} \quad \boxed{B = \gamma P} \quad \text{and} \quad \boxed{v = \sqrt{\frac{\gamma P}{\rho}}}$$

Now we can use the ideal gas law

$$PV = nRT \Rightarrow P = \frac{n}{V}RT = \frac{\rho}{M}RT$$ where M = mean molecular mass

Therefore we can rewrite the wave speed as:

$$v = \sqrt{\frac{\gamma RT}{M}}$$

T = temp in Kelvin

$R = 8.314\ J/mol \cdot K$

M = molecular mass (kg/mol)

$\gamma = \begin{cases} 5/3 & \text{monatomic gas} \\ 7/5 & \text{diatomic gas} \end{cases}$

Example1: speed of sound in ^4He at 20°C

^4He has 2 protons and 2 neutrons

$$\Rightarrow M = 4.00\ g/mol = 4\times10^{-3}\ kg/mol$$

$$T = 273 + 20 = 293K$$

^4He is monatomic so $\gamma = 5/3$

$$v = \sqrt{\frac{(5/3)(8.314\ J/mol\cdot K)(293K)}{(4\times10^{-3}\ kg/mol)}} = \boxed{1010\ m/s}$$

Example 2: speed of sound in air (80% N_2, 20% O_2) at 20°C

Atomic masses: $N_2 = 14.0\,u$ (7 protons, 7 neutrons)

$O_2 = 16.0\,u$ (8 protons, 8 neutrons)

$M = 0.2(2 \times 16.0\ g/mol) + 0.8(2 \times 14.0\ g/mol) = 28.8\ g/mol$
$= 2.88 \times 10^{-2}\ kg/mol$

Both gases are diatomic, so $\gamma = 7/5$

$T = 273 + 20 = 293\ K$

$$\Rightarrow v = \sqrt{\frac{(7/5)(8.314\ J/mol \cdot K)(293\ K)}{(2.88 \times 10^{-2}\ kg/mol)}} = \boxed{344\ m/s}$$